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TCAD-Net: Integrating Transformer and Convolution for Weakly Supervised Anomaly Data Discovery

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Abstract*—With the continuous development of consumer electronic device functions and the large-scale data, ensuring device safety and privacy has become increasingly important. Deep learning-based weakly supervised anomaly detection methods have been widely applied in various fields such as consumer finance, significantly improving the accuracy and reliability of discovery. However, existing methods typically either utilize convolution or Transformer, thus unable to effectively leverage both the local features and long-range dependencies of the large-scale samples simultaneously. To address this challenge, this work proposes a novel deep learning-based weakly supervised anomaly detection network, named TCAD-Net, which effectively integrates the advantages of convolution and Transformer architectures, thus possessing efficient anomaly data discovery capability. Specifically, TCAD-Net is a dual-channel framework, where the convolutional shrinkage-based feature extraction branch and the Transformer-based feature extraction branch are used for extracting local features and long-range dependencies of the large-scale samples, respectively. Subsequently, these two branches are merged through a bridging unit and then fed into an anomaly score generator module unit to obtain the anomaly scores of samples. Extensive experiments demonstrate that the proposed TCAD-Net exhibits exceptional performance across various large-scale benchmark datasets. Its average performance in terms of Precision-Recall curve area under the curve and Receiver Operating Characteristic curve area under the curve surpasses that of state-of-the-art methods by 11.74% and 4.43%, respectively. The codes and trained models are released at TCAD-Net.

***Index Terms*—Consumer electronics, Anomaly data discovery, Transformer, Imbalanced samples, Large-scale data**

I. INTRODUCTION

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THE widespread adoption of consumer electronics has fundamentally transformed modern lifestyles [1], [2]. However, this transformation has introduced significant challenges in ensuring the reliability and security of large-scale consumer data. Anomaly data discovery has emerged as a critical technology for addressing these challenges. This technology focuses on identifying samples within datasets that deviate from the majority or exhibit unexpected behavioral patterns, thereby enhancing the overall reliability and security of consumer data. For example, in fields such as consumer electronics, financial fraud, and communication security [3]–[7], anomalies can manifest as device malfunctions, security breaches, or abnormal user behaviors, respectively. Therefore, effective anomaly detection plays a crucial role in uncovering potential issues or irregularities, enabling timely intervention or adjustments.

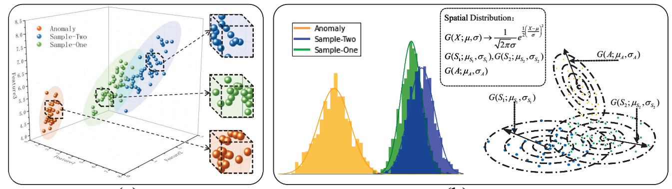
Traditional anomaly data discovery methods typically rely on rule-based systems or probabilistic statistical approaches [8], [9]. However, these methods often have limitations, including insufficient adaptability, high false alarm rates, low efficiency in processing large-scale data, and difficulty in detecting novel anomalies, thus failing to effectively handle complex, dynamic data patterns. With the continuous advancement of Deep Learning (DL) technology, DL-based anomaly data discovery can capture subtle and intricate patterns indicating anomalies from large-scale benchmark datasets [10], [11]. Depending on the ratio of normal to anomalous samples during model training, DL-based anomaly detection methods can be further divided into supervised, unsupervised, and weakly supervised techniques. Specifically, supervised learning often exhibits high accuracy and robustness in anomaly detection. However, it also imposes stringent requirements on the balance of labeled samples, with minority class samples often being scarce or costly. Unsupervised learning, on the other hand, can train models using single-class labels, achieving anomaly data discovery by identifying data that significantly deviates from the training samples. Although unsupervised learning overcomes the dependency on minority class samples, the training data often contain noise or inaccuracies due to manual labeling errors, leading to an increase in false positives or false negatives.

Based on the characteristics of supervised and unsupervised learning methods, each is suitable for different application scenarios, making the choice of the appropriate method particularly important. Supervised learning is suitable for scenarios with abundant labeled data and a need for high-precision predictions, while unsupervised learning is employed to uncover the inherent structures of data and in situations where data labeling is challenging. Therefore, weakly supervised learning, which balances learning efficiency and performance with limited labeled data, has emerged as an important research direction that lies between supervised and unsupervised learning.

Weakly supervised learning anomaly data discovery methods [12]–[14] commonly employ prevalent DL techniques, such as Multilayer Perceptrons (MLP), pure Convolutional Networks (ConvNets), for feature extraction from imbalanced samples. These methods address the challenge of having only a few anomalous samples and noisy data during the training phase by enhancing the learning weights of minority class samples or incorporating prior knowledge of the samples, thereby enabling effective anomaly data discovery. Consequently, such anomaly data discovery methods successfully avoid the dependence on a large number of annotated samples while improving the performance and accuracy of anomaly detection in imbalanced data scenarios. For instance, Guan et al. [15] proposed WAKE, which combines auto-encoders and Multilayer Perceptrons to achieve anomaly data discovery in log data. Meanwhile, given the outstanding performance of Transformers in modeling long-range dependencies [16]–[19], researchers have initiated efforts to apply Transformer architectures to the discovery of anomaly data across diverse anomaly data discovery tasks. However, existing methods still exhibit the following limitations on large-scale benchmark datasets:

- Weakly supervised anomaly data discovery methods based on MLPs or ConvNets face limitations in effectively capturing long-range dependencies within large-scale samples. These limitations primarily stem from restricted receptive fields or the use of fixed-length hidden representations.
- Transformer-based weakly supervised anomaly data discovery methods may lack the inherent inductive biases of ConvNets, which leads to suboptimal performance in capturing local features for anomaly detection tasks.

Fig. 1 illustrates a typical data distribution. By combining the two subfigures, one can gain a comprehensive understanding of the local and long-range dependencies of sample features. Fig. 1(a) illustrates the local feature dependencies among samples within a smaller range, while Fig. 1(b) reveals the long-range dependencies among samples at a larger spatial scale. Typically, the local and long-range dependencies between samples can influence the low-dimensional space projection of the data, allowing it to be observed in the low-dimensional space [20]. This naturally raises the question: Can one design an anomaly data discovery method that can simultaneously extract both local features and long-range dependencies of the data?



(a) (b)

Fig. 1: Schematic representation of feature space distribution: (a) represents the correlation of local features among samples; (b) depicts the low-dimensional projection of long-range dependencies between samples, indicating that the low-dimensional projections of high-dimensional samples exhibit a quasi-Gaussian distribution.

A. Motivation and Contributions

To address this question, this work proposes TCAD-Net, a dual-branch weakly supervised anomaly detection network, which possesses efficient anomaly data discovery capabilities. Specifically, the Feature Extraction Module (FEM) of TCAD-Net consists of two branches: the Transformer-based Feature extraction Branch (TFB) and the Convolutional Shrinkage-based feature extraction Branch (CSB). TFB is primarily responsible for capturing the long-range dependencies representations of samples, while CSB is primarily used to extract the local features of samples. Moreover, CSB enhances the local feature representation of samples through techniques such as convolution, down-sampling, and average pooling, rather than simply stacking convolutional layers. Additionally, a weighted soft-thresholding mechanism in CSB is introduced to compress and filter local features, thereby eliminating insignificant features. To facilitate the representation of sample features by both the Transformer and ConvNets, the outputs of the two feature extraction branches are integrated through a bridging unit. Finally, the integrated features are input into an Anomaly Score Generator Module (ASGM) to calculate the anomaly score of the samples. As a result, TCAD-Net efficiently integrates the advantages of convolution and Transformer architectures to achieve accurate and reliable weakly supervised anomaly data discovery. The main contributions of this paper can be summarized as follows:

- We propose a novel deep learning based weakly supervised anomaly data discovery network, dubbed as TCAD-Net. It integrates the advantages of Transformer and ConvNet architectures, effectively capturing local features and long-range dependencies of samples to achieve more accurate anomaly data discovery.
- We develop TFB for TCAD-Net that captures the long-range dependencies representation of samples, which can mitigate the inherent restrictions of the receptive field in ConvNets.
- We design CSB for TCAD-Net that obtains local features of data, which can reduce the impact of insignificant features on anomaly data discovery results by introducing a weighted soft-thresholding operator.
- We introduce a feature integration module for TCAD-Net that facilitates the fusion of features from TFB and CSB, enhancing the accuracy of anomaly discovery.

Experimental results indicate that, in terms of Precision-Recall curve Area Under the Curve (PR-AUC), Receiver Operating Characteristic curve Area Under the Curve (ROC-AUC) and

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F1-Score, TCAD-Net outperforms other state-of-the-art methods significantly on eight large-scale benchmark datasets.

B. Organization

The remainder of this paper is structured as follows: Section II reviews the latest research and advancements in anomaly detection methods. Section III provides a detailed explanation of the proposed TCAD-Net framework, breaking it down from various key dimensions, including its core architecture, technical principles, and algorithmic optimizations. Section IV describes the datasets, parameter settings, and experimental results. Finally, Sections V and VI present the ablation studies related to TCAD-Net and summarize the findings of this research.

C. Abbreviations and Notations

Table I presents the meanings of the abbreviations and notations used in TCAD-Net. This table serves as a reference to ensure clear and consistent interpretation of the various terms and symbols used in the model description, architecture, and experimental analysis.

TABLE I: Abbreviations and notations in TCAD-Net.

Abbreviations and notations	Meaning
FEM	Feature Extraction Module
ASGM	Anomaly Score Generator Module
TFB	Transformer-based Feature Extraction Branch
CSB	Convolutional Shrinkage-based Feature Extraction Branch
LConvs	Lightweight Convolutional
RS	Residual Shrinkage
$\boldsymbol{x}_{tfb}$	Output of TFB
$\boldsymbol{x}_{csb}$	Output of CSB
$\boldsymbol{x}_{con}$	Output of Feature Fusion of $\boldsymbol{x}_{tfb}$ and $\boldsymbol{x}_{csb}$
$\hat{y}$	Anomaly Score
$\boldsymbol{x}$	Input Sample
B	Batch Size
D	Feature Dimension of $\boldsymbol{x}_{tfb}$
D	Feature Dimension of $\boldsymbol{x}_{csb}$
p	Positional Encoding Blocks
M	Number of Iterations for TFB
N	Number of Iterations for CSB
α	Balance Factor of Loss Function
γ	Adjustable Multiplier of Loss Function
$\mathcal{M}_{Re}(\cdot)$	Dimension-raising Matrix
$\mathcal{M}_{Eb}(\cdot)$	Position Encoding Matrix
$\mathcal{LN}(\cdot)$	Layer Normalization
$\mathcal{G}(\cdot)$	Sigmoid Activation Function
$\mathcal{R}(\cdot)$	ReLU Activation Function
$Conv(\cdot)$	Convolution
$\mathcal{BN}(\cdot)$	Batch Normalization
$\mathcal{F}_{con}(\cdot)$	Feature Fusionmatrix
$Dropout(\cdot)$	Dropout Function
$\mathcal{L}_{\text{TCAD}}(\cdot)$	Loss Function

II. Related Works

A. Supervised Anomaly Data Discovery Methods

Supervised anomaly data discovery relies on learning the distribution patterns of data from a well-labeled training dataset to achieve accurate data discovery. These methods primarily utilize ConvNets [21], auto-encoders [10], and Long Short-Term Memory (LSTM) [11] to extract data features for identifying anomalous data. For example, Heng et al. [21] introduced a CNN-LSTM model with attention, for intrusion detection in controller area network, utilizing Conv1D for feature extraction and attention-based Bi-LSTM for temporal dependency analysis. Similarly, Liu et al. [11] proposed a supervised network integrating LSTM and ConvNets for anomaly detection in data packet payloads. Furthermore, Deng et al. [22] proposed a spatio-temporal graph convolutional adversarial network, employing spatiotemporal generators and discriminators to predict normal traffic dynamics and detect anomalies, achieving robust data discovery through novel anomaly scores. Hojjati et al. [10] proposed a one-class anomaly data discovery method, which exhibits good performance in anomaly data detection.

Supervised anomaly data discovery methods often achieve satisfactory detection performance when there is an abundance of balanced sample data. However, in practical applications, challenges such as scarce and imbalanced sample data, as well as high annotation costs, are commonly encountered, which typically limit the application of supervised learning methods.

B. Unsupervised Anomaly Data Discovery Methods

Unsupervised anomaly data discovery methods utilize the input data itself as the supervisory signal to guide the model in learning the one-to-one mapping relationship of samples, resulting in a reconstruction output. These methods can analyze the intrinsic structure and features of the data to identify anomalous samples without relying on balanced annotated training data. Through techniques such as Isolation Forest (iForest) [23], RNN [24], principal component analysis [25], and others, these methods can automatically learn behavioral patterns from singly labeled data and identify outlier data points deviating from these patterns.

For example, Liu et al. [23] introduced iForest, which does not rely on distance or density metrics, enabling effective anomaly data discovery. Similarly, Xiang et al. [26] proposed Deep-iForest, a deep anomaly detection framework based on hash isolation forests, which enhances feature representations using isolation forests and tree embedding schemes, achieving accurate anomaly detection while preserving the advantages of deep forests. Additionally, Li et al. [27] introduced COPOD, which computes p-values for each anomaly by modeling multivariate data distributions, determining their tail probabilities. Subsequently, inspired by the notion that anomalies are “rare events” in distribution tails, Li et al. [28] proposed the ECOD algorithm, which models the empirical cumulative distribution for each dimension to estimate the underlying distribution of the input data. In contrast, Luo et al. [24] introduced the stacked RNN, which utilizes temporal coherent sparse coding to preserve inter-frame similarity, enabling real-time anomaly data discovery.

Unsupervised anomaly data discovery methods can effectively discover anomalous patterns in cases of missing samples or incomplete labeling, exhibiting strong anomaly detection performance. However, in scenarios where training samples contain imbalanced data, noise, or inaccuracies or errors in manual labeling, unsupervised learning methods sensitive to training data can often lead to a high number of false positives or false negatives in detection results.

C. Weakly Supervised Anomaly Data Discovery Methods

Due to the limitations of both supervised and unsupervised learning methods for anomaly detection, researchers have increasingly focused on weakly supervised anomaly data discovery. These approaches employ imbalanced labeled samples to train models and tackle the highly imbalanced distributions between normal and anomaly samples by employing techniques such as incorporating sample priors, optimizing loss functions, and pretraining during the training process. Some notable anomaly data discovery algorithms in this category include FSNet [29], REPEN [30], DevNet [31], Deep SAD [32], FEWAE [33], and Dual-MGAN [34].

Specifically, Jack et al. [29] introduced the prototype network, named FSNet, which performs classification and anomaly detection by computing distances between sample prototype representations. Subsequently, Pang et al. [30] designed REPEN , which employs a ranking model for anomaly data discovery. REPEN enhances the performance of anomaly data discovery by detecting outliers. Following this work, Pang et al. [31] proposed DevNet, which integrates prior data information into the model training process to guide optimization, achieving end-to-end anomaly sample identification. Building upon the DevNet method, Zhou et al. [33] developed FEWAE, which pretrains input data using auto-encoders to acquire high-dimensional representations, followed by combining MLP to obtain sample anomaly scores. Additionally, Ruff et al. [32] introduced Deep-SAD, which models training samples using auto-encoders and adjusts the entropy of potential distributions of normal and anomaly samples using information entropy to achieve accurate weakly supervised anomaly data discovery. Li et al. [34] proposed Dual-MGAN based on GANs, decomposing the detection task into two parts: the distribution construction module and the data enhancement module. Meanwhile, Xu et al. [16] introduced a weakly supervised semantic segmentation framework based on the Transformer, utilizing a contrastive-class-token module to generate precise class-specific object localization maps.

TABLE II: Comparison of the key features of TCAD-Net with other competing methods.

Algorithm	Key Components	Key Technical Principles	Local Features	Long-range Dependencies
iForest	Isolation Tree	Decision Tree	NA	NA
Copod	Data Distribution	Abnormal p-value	NA	NA
Deep-iForest	Isolation Tree	Tree-embedding	NA	NA
Ecod	Data Distribution	Tail Probabilities Across Dimensions	NA	NA
FSNet	MLP	Distance Measure	Yes	NA
REPEN	MLP	Low-dimensional Projection	Yes	NA
DevNet	MLP	Prior Knowledge	Yes	NA
Deep-SAD	Autoencoder	Information Entropy	Yes	NA
FEWAE	Autoencoder	High-dimensional Representations	Yes	NA
Dual-MGAN	GAN	Distribution Construction	Yes	NA
TCAD-Net	Transformer+CNN	Dual-channel Framework	Yes	Yes

These weakly supervised learning methods for anomaly data discovery effectively alleviate the reliance on balanced samples, while enhancing the performance and accuracy of anomaly detection in the presence of noise. However, existing methods typically either utilize MLP or ConvNets or autoencoder, thus unable to effectively leverage both the local features and long-range dependencies in complex and high-dimensional large-scale data distribution scenarios simultaneously. As shown in Table II, the proposed dual-channel architecture of TCAD-Net integrates a Transformer-based branch for capturing long-range dependencies and a convolution-based branch for enhancing local features. Through feature integration techniques and an anomaly score generator, it enables collaborative processing of both local and global sample features, thereby achieving more precise anomaly detection.

III. Architecture of TCAD-Net

A. Overview of TCAD-Net

Fig. 2 illustrates the overall architecture of TCAD-Net. TCAD-Net primarily consists of two key modules: the Feature Extraction Module (FEM) and the Anomaly Score Generation Module (ASGM). Specifically, the FEM consists of two main feature extraction branches: the Transformer-based Feature extraction Branch (TFB) and the Convolutional Shrinkage-based feature extraction Branch (CSB). Subsequently, the outputs of TFB and CSB are integrated and fed into the ASGM to obtain the anomaly scores of the samples.

B. Feature Extraction Module

The FEM comprises TFB and CSB, which are used to extract both global and local feature information from the input data at different levels and dimensions. Specifically, the TFB consists of multi-head attention layers and Lightweight Convolutional (LConvs) layers. This module captures long-range dependencies information from the input samples while integrating long-range dependencies representations through the LConvs layers. The CSB is composed of convolutional layers and Residual Shrinkage (RS) layers, which filter out insignificant features by calculating the weighted soft threshold of the samples, thereby achieving more effective feature learning.

1) TFB: This branch is used to extract long-range dependencies representations of the samples, which is composed of $M = 8$ identical sub-layers in series. It takes the samples $\boldsymbol{x}$ as input, where $\boldsymbol{x} \in \mathbb{R}^{B*D}$ is divided into batches of size B, and D denotes the feature dimension of $\boldsymbol{x}$. Subsequently, $\boldsymbol{x}$ is mapped into high-dimensional feature representations with positional information through positional encoding.

Specifically, the positional information is embeded into the samples $\boldsymbol{x}$ by constructing a positional encoding matrix $\mathcal{P}$ (with dimensions $\mathbb{R}^{B*p}$, where p represents the number of positional encoding blocks). TFB discards the traditional method of generating positional encoding matrices using sine and cosine functions. Instead, it encodes the input sample $\boldsymbol{x}$ using an orderly positional encoding matrix $\mathcal{P}$, simplifying the encoding process while mapping low-dimensional features to high-dimensional representations. As a result, the encoded $\boldsymbol{x}$ is transformed from a size of $\mathbb{R}^{B*D}$ to a high-dimensional feature matrix of size $\mathbb{R}^{B*p*D}$. This process can be formalized as follows:

$$\bar{\boldsymbol{x}} = \mathcal{M}_{Re}(\boldsymbol{x}) + \mathcal{M}_{Eb}(\mathcal{P}) \tag{1}$$

where $\bar{\boldsymbol{x}}$ represents the feature matrix with position information added, $\mathcal{M}_{Eb}(\mathcal{P})$ represents the position encoding matrix, $\mathcal{M}_{Re}$ represents a dimension-raising matrix to adjust the feature dimensions of $\boldsymbol{x}$ so that it can match the dimensions of $\mathcal{P}$.

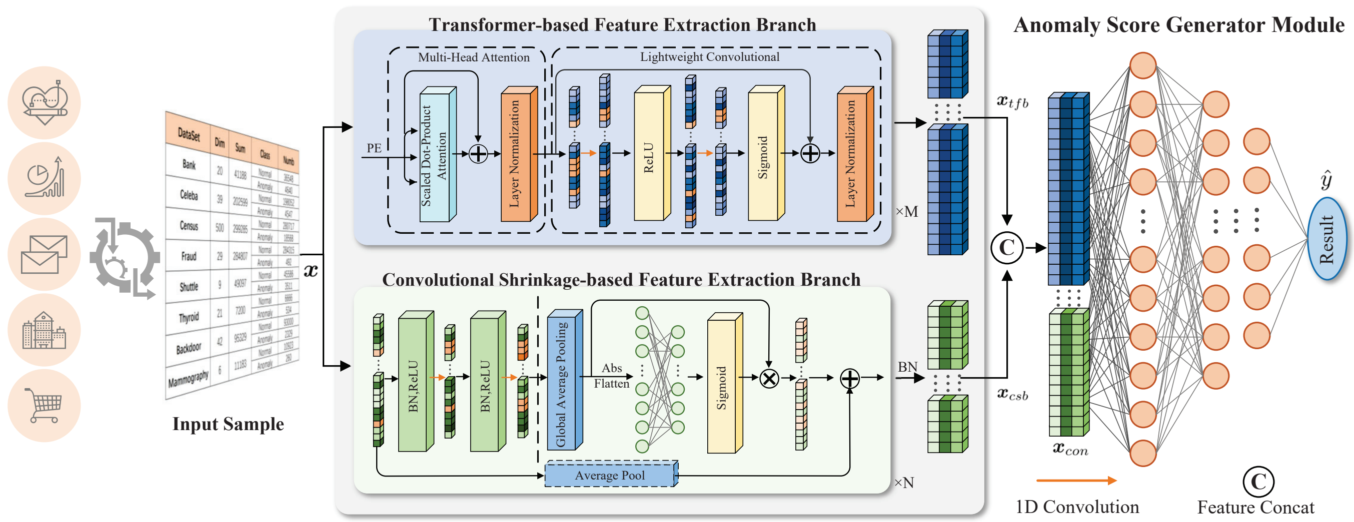

Fig. 2: The proposed TCAD-Net consists of a Feature Extraction Module (FEM) and an Anomaly Score Generation Module (ASGM). The FEM includes two main feature extraction branches: TFB and CSB. The symbol M denotes the number of iterations in the TFB, while symbol N represents the number of iterations in the CSB.

Subsequently, the feature matrix $\bar{\boldsymbol{x}}$ is fed into the multi-head attention module, where it undergoes linear projections to obtain three linear matrices: $\boldsymbol{Q}$ (query), $\boldsymbol{K}$ (key), and $\boldsymbol{V}$ (value). These matrices form the basis for operations within the multi-head attention layer. Formally, one can obtain:

$$\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i = \mathcal{F}_{\boldsymbol{QKV}}(\bar{\boldsymbol{x}}, D, h) \quad (i \in \{1, \cdots, h\}) \tag{2}$$

where $\mathcal{F}_{\boldsymbol{QKV}}$ denotes the linear projection of $\bar{\boldsymbol{x}}$. As a result, $\bar{\boldsymbol{x}}$ is projected into h (a factor of the input data dimension) sets of independent queries, keys, and values, denoted as $\boldsymbol{Q}_i$, $\boldsymbol{K}_i$, and $\boldsymbol{V}_i$ $\left(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i \in \mathbb{R}^{B*p*h*\frac{D}{h}}\right)$. Each set is used to compute attention scores, which determine the relevance of other tokens for a particular token. Additionally, in the multi-head attention layer, scaled dot-product attention is used to compute the similarity between different positions of the input encoded samples. It is important to note that multi-head attention includes h scaled dot-product attentions, and their outputs are concatenated along the dimension, resulting in a size of $\mathbb{R}^{B*p*D}$. Specifically, this is achieved through matrix operations on $\boldsymbol{Q}_i$, K_i, and $\boldsymbol{V}_i$:

$$\begin{cases} H_i = Attention\left(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i\right) \quad (i \in \{1, \cdots, h\}) \\ Attention(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i) = Softmax(\frac{\boldsymbol{Q}_i \cdot \boldsymbol{K}_i^T}{\sqrt{D}}) \cdot \boldsymbol{V}_i \\ \mathrm{MHA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathcal{F}_{proj}\left(H_1, \cdots, H_h\right) \end{cases} \tag{3}$$

where the dot product of $\boldsymbol{Q}_i$ and $\boldsymbol{K}_i$ is scaled by the square root of the key's dimension D. Next, the softmax function is applied to compute the attention weights for matrix $\boldsymbol{V}_i$. $\mathcal{F}_{proj}$ concatenates H_i along the specified dimension and linearly projects the result to the feature dimension. Multi-head attention layer has been demonstrated to capture dependencies among feature matrices and enhance the model's generalization ability. Subsequently, the output of multi-head attention layer is normalized using layer normalization, as follows:

$$\boldsymbol{x}' = \mathcal{LN}(\mathrm{MHA}(\bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}) + \bar{\boldsymbol{x}}) \tag{4}$$

where $\mathrm{MHA}(\bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}, \bar{\boldsymbol{x}})$ represents the feature matrix $\bar{\boldsymbol{x}}$ as the input of the multi-head attention layer, with the symbol "+" representing matrix addition, $\mathcal{LN}(\cdot)$ denotes the layer normalization operation with the parameter $\omega_{\mathcal{LN}}$, $\boldsymbol{x}'$ represents the output of the multi-head attention layer after normalization. Subsequently, the features of the samples are extracted through a convolution operation, as expressed in the following equation:

$$Conv(\boldsymbol{x}') = \sum_{j=0}^{k-1} W_j \bigotimes \boldsymbol{x}' + b \tag{5}$$

where $Conv(\boldsymbol{x}')$ represents the output after $\boldsymbol{x}'$ passes through the convolution; W_j denotes the j-th weight parameter of the convolutional kernel; b represents the bias term; k denotes the size of the convolutional kernel. Therefore, the 1-st iteration LConvs layer output can be expressed as:

$$\begin{cases} \boldsymbol{x}'' = \mathcal{G}(Conv(\mathcal{R}((Conv(\boldsymbol{x}'))))) \\ \boldsymbol{x}_{(tfb,1)} = \mathcal{LN}(\boldsymbol{x}'' + \boldsymbol{x}') \end{cases} \tag{6}$$

where $\mathcal{G}(\cdot)$ and $\mathcal{R}(\cdot)$ represent Sigmoid and ReLU activation functions, respectively; $\left\{\boldsymbol{x}'' | \boldsymbol{x}'' \in \mathbb{R}^{B*p*D}\right\}$ represents the output of the two convolutions. Subsequently, $\boldsymbol{x}''$ is normalized, resulting in the output of the 1-st layer of TFB, denoted as $\boldsymbol{x}_{(tfb,1)}$. It is worth noting that the output $\boldsymbol{x}_{(tfb,1)}$ has the same dimensions as the input $\boldsymbol{x}$. The iteration depth of TFB is set to $M = 8$ layers to enable it to capture deep-level long-range dependencies. That is, TFB is composed of $M = 8$ sub-layers with consistent input and output dimensions.Therefore, the final output formula of TFB is expressed as:

$$\boldsymbol{x}_{tfb} = \boldsymbol{x}_{(tfb,M)} = \mathcal{LN}\left(\boldsymbol{x}''_{(tfb,M-1)} + \boldsymbol{x}'_{(tfb,M-1)}\right) \tag{7}$$

where $\boldsymbol{x}_{(tfb,M)}$ is abbreviated as $\boldsymbol{x}_{tfb}$. Note that $\mathcal{LN}(\cdot)$ is used to normalize the output of the TFB, ensuring a more stable input data distribution across layers. Overall, the TFB

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Algorithm 1 Feature Extraction Module

Input: $\{\boldsymbol{x}|\boldsymbol{x} \in \mathbb{R}^{B*D}\}$
Output: $\{\boldsymbol{x}_{tfb}|\boldsymbol{x}_{tfb} \in \mathbb{R}^{B*p*D}\}$; $\{\boldsymbol{x}_{csb}|\boldsymbol{x}_{csb} \in \mathbb{R}^{B*p*S}\}$

```
Initialization:
    Initialize the number of iterations for TFB to M, position encoding dimension to p and the number of heads to h;
    Initialize the number of iterations for CSB to N, kernel size k, stride s; RS layers parameters: Average Pooling down_sap
Workflow of TFB:
Calculate the feature matrix x̄ by Eq. (1)
for m = 1 to M do
    Multi-Head Attention Layer:
    Q_i, K_i, V_i ← F_QKV(x̄, D, h)
    Calculate the output x' of the multi-head attention layer using Q_i, K_i, V_i according to Eq. (3) and (4).
    LConvs Layer:
    x'' ← G(Conv(R((Conv(x')))))
    Calculate the output x_(tfb,m) of LConvs Layer according to Eq. (6) and Eq. (7)
end for
Output1: x_tfb ← x_(tfb,M)
Workflow of CSB:
x̂ = M_Re(x)(x ∈ R^{B*C_in*D})          ▷ Reshape the input matrix
for n = 1 to N do
    Convolutional Layers:
    Pass the x̂ through convolutional layers
    Calculate the output x_(Covs,n) of convolutional layers according to Eq. (5) and Eq. (9)
    RS Layers:
    Compute the weighted soft thresholds x_(softh,n) for RS layers according to Eq. (10)
    Store the combined features in output_features
    if down_sap is True then
        x̂ = Pool_ap(x̂)                   ▷ Using Eq. (10)
    end if
    x_(rsl,n) ← x_(softh,n) + x̂           ▷ Residual Connection
    x̂ ← x_(rsl,n)
end for
Output2: x_csb ← BN(x_(rsl,N))
```

effectively leverages the Transformer's capability to capture long-range dependencies among samples, thereby generating high-dimensional feature representations of the aggregated samples, denoted as $\boldsymbol{x}_{tfb}$.

2) CSB: By integrating convolutional layers and RS layers, CSB extracts local information at different levels from the raw sample $\boldsymbol{x}$. CSB is composed of $N = 7$ sub-layers with the same structure, and the final output is further enhanced by Batch Normalization $\mathcal{BN}(\cdot)$ to strengthen the expression of local information. Specifically, the RS layer applies pooling, Fully Connected Layer (FCL), and residual connections to compute the weighted soft threshold for the input samples. This weighted soft threshold function compresses and filters the features based on their importance, effectively removing insignificant features for more efficient feature learning.

Firstly, CSB introduce $\mathcal{BN}(\cdot)$ to ensure that the sample features have zero mean and unit variance. By calculating the mean μ_{bt} and standard deviation σ_{bt} of sample $\boldsymbol{x}$ in each mini-batch, one can obtain:

$$\begin{cases} \mu_{bt} = \frac{1}{N_{bt}} \sum_{n=1}^{N_{bt}} \boldsymbol{x} \\ \sigma_{bt}^2 = \frac{1}{N_{bt}} \sum_{n=1}^{N_{bt}} (\boldsymbol{x} - \mu_{bt})^2 \\ \mathcal{BN}(\boldsymbol{x}) = \gamma \frac{\boldsymbol{x} - \mu_{bt}}{\sqrt{\sigma_{bt}^2 + \epsilon}} + \beta \end{cases} \tag{8}$$

where N_{bt} denotes the size of the mini-batch, and ϵ is a small constant added to prevent division by zero during computation. Subsequently, the output is adjusted using two trainable parameters, γ (scale factor) and β (shift parameter). Subsequently, $\mathcal{R}$ is used as the activation function for the output. Thus, the output of the 1-st convolutional layer can be represented as:

$$\begin{cases} \boldsymbol{x}_{(mid,1)} = Conv(\mathcal{R}(\mathcal{BN}(\mathcal{M}_{Re}(\boldsymbol{x})))) \\ \boldsymbol{x}_{(Covs,1)} = Conv(\mathcal{R}(\mathcal{BN}(\boldsymbol{x}_{(mid,1)}))) \end{cases} \tag{9}$$

where $\boldsymbol{x}_{(Covs,1)}$ represents the output of the 1-st convolutional layer with dimensions $\mathbb{R}^{B*C_{out}^1*S}$ (C_{out}^1 represents the output intermediate dimension; $S \approx \frac{D}{2}$); $\boldsymbol{x}_{(mid,1)}$ represents the output of the middle layer; $\boldsymbol{x}_{(Covs,1)}$ is then passed through the RS layer to obtain the weighted soft threshold, which compresses the non-significant features in the sample to values close to zero, enabling the network to focus more on significant features.

Specifically, the Global Average Pooling (GAP), dimensionality reduction, and FCL are introduced to map the features of $\boldsymbol{x}_{(Covs,1)}$, and enhance the expression of input sample features by constructing residual blocks. This process can be formulated as follows:

$$\begin{cases} \boldsymbol{x}_{(Covs,1)}^{'} = \mathcal{F}_{flat}\left(Pool_{gap}\left(\left|\boldsymbol{x}_{(Covs,1)}\right|\right)\right) \\ \eta = \mathcal{G}\left(\mathcal{F}_{FCL}\left(\boldsymbol{x}_{(Covs,1)}^{'}\right)\right) \\ \boldsymbol{x}_{(softh,1)} = \Phi * max\left(0, \left|\boldsymbol{x}_{(Covs,1)}\right| - \eta \boldsymbol{x}_{(Covs,1)}^{'}\right) \\ \boldsymbol{x}_{(rsl,1)} = \boldsymbol{x}_{(softh,1)} + Pool_{ap}(\boldsymbol{x}) \end{cases} \tag{10}$$

where $Pool_{gap}(\cdot)$ represents the GAP; $\mathcal{F}_{flat}(\cdot)$ is the function for dimensionality reduction of multi-dimensional data; η represents the output of $\boldsymbol{x}_{(Covs,1)}$ after passing through FCL and sigmoid; $\boldsymbol{x}_{softh}$ denotes the weighted soft threshold obtained from the RS layer for the input $\boldsymbol{x}_{(Covs,1)}$, with the soft threshold being a non-negative value; Φ represents the adjustment weight of the soft threshold, and the soft threshold is summed element-wise with the pooled input sample $Pool_{ap}(\cdot)$; $\boldsymbol{x}_{(rsl,1)}$ represents the output of the 1-st RS layer, with a size of $\mathbb{R}^{B*C_{out}^1*S}$. Note that $Pool_{ap}(\cdot)$ is only applied in the 1-st RS layer, and subsequent iterative layers use the output of the previous layer for summation. Formally, one can obtain:

$$\boldsymbol{x}_{(rsl,j)} = \boldsymbol{x}_{(softh,j-1)} + \boldsymbol{x}_{(rsl,j-1)} (j \in \{2, \cdots, N\}) \tag{11}$$

where $\left\{\boldsymbol{x}_{(rsl,j)}|\boldsymbol{x}_{(rsl,j)} \in \mathbb{R}^{B*C_{out}^j*S}\right\}$ denotes the output of the j-th RS layer. Note that too many iterations in the CSB increase the model's complexity, while too few iterations reduce

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its expressive capability. Therefore, CSB employs $N = 7$ sub-layers with the same structure for iteration, and combines the output of the final layer using batch normalization as described in Eq. 8. The output expression of CSB is given by:

$$\boldsymbol{x}_{csb} = \mathcal{BN}\left(\boldsymbol{x}_{(rsl,N)}\right) \tag{12}$$

where $\boldsymbol{x}_{(rsl,N)}$ represents the output of the N-th RS layer. $\left\{\boldsymbol{x}_{csb} | \boldsymbol{x}_{csb} \in \mathbb{R}^{B*C_{out}^N*S}\right\}$ denotes the output of $\boldsymbol{x}_{(rsl,N)}$ after passing through the batch normalization, where $C_{out}^N = p$. $\boldsymbol{x}_{csb}$ signifies the local feature representation extracted from the feature data after passing through CSB, effectively filtering out insignificant features within the samples.

Alg.1 illustrates the comprehensive workflow of the FEM. In general, through FEM, one can obtain the matrices $\boldsymbol{x}_{ftb}$ and $\boldsymbol{x}_{csb}$, which encompass both macro long-range dependency information and micro local feature representations of samples. Due to their consistent spatial dimensions,$\boldsymbol{x}_{ftb}$ and $\boldsymbol{x}_{csb}$ can be merged along the feature dimension. Notably, compared to the original samples, the representations obtained through FEM, $\boldsymbol{x}_{ftb}$ and $\boldsymbol{x}_{csb}$, exhibit higher dimensionality, enabling them to compensate for the limitations of individual features.

C. Anomaly Score Generator Module

The ASGM is capable of modeling complex and aggregated high-dimensional feature representations to obtain the anomaly scores of input samples. Before obtaining the anomaly scores, the outputs of TFB and CSB, $\boldsymbol{x}_{tfb}$ and $\boldsymbol{x}_{csb}$, need to be merged along the feature dimension. As described before, $\boldsymbol{x}_{tfb} \in \mathbb{R}^{B*p*D}$, where p represents the number of encoding blocks and D represents the feature dimension, and $\boldsymbol{x}_{csb} \in \mathbb{R}^{B*p*S}$, where S represents the output feature dimension of the CSB. Notably, CSB compresses the input sample along the feature dimension once through average pooling. If multiple compressions are performed on the input, S would be less than $\frac{D}{2}$. Let $\left\{\boldsymbol{x}_{con} | \boldsymbol{x}_{con} \in \mathbb{R}^{B*p*(D+S)}\right\}$ denote the combined feature. As a result, $\boldsymbol{x}_{con}$ can be expressed as:

$$\boldsymbol{x}_{con} = \mathcal{F}_{con}(\boldsymbol{x}_{tfb}, \boldsymbol{x}_{csb}) \tag{13}$$

where $\mathcal{F}_{con}(\cdot)$ represents the concatenation operation along the feature dimension. By concatenating features, this operation preserves multidimensional information while effectively mitigating the risk of significant information loss caused by excessive compression. Finally, $\boldsymbol{x}_{con}$ is fed into the ASGM to obtain the anomaly score $\hat{y}$. This score represents the probability of the input sample being anomalous, as described by the following formula:

$$\begin{cases} \boldsymbol{z}_1 = \mathcal{R}(W_1\mathcal{F}_{flat}(\boldsymbol{x}_{con}) + b_1) \\ \boldsymbol{z}_l = \mathcal{R}(W_l\boldsymbol{z}_{l-1} + b_l) \quad (l \in \{1, \cdots, L\}) \\ \hat{y} = Dropout(\mathcal{G}(W_L\boldsymbol{z}_{L-1} + b_L)) \end{cases} \tag{14}$$

where W represents the connection coefficients; b represents the bias; L denotes the number of layers, $Dropout(\cdot)$ denotes the dropout function to prevent overfitting, and $\hat{y}$ represents the anomaly score of the sample.

D. Loss Function

In the context of weakly supervised anomaly data discovery, sample imbalance significantly impacts the accuracy and reliability of the model. Therefore, the Focal loss [35] is used as the loss function for TCAD-Net. This function adjusts the model's training direction through weighting, aiming to optimize and enhance the model's performance. It introduces a modulation factor to reduce the loss weight for easily classified samples and increase it for difficult-to-classify samples. This strategy guides the model to better learn the minority class samples that are often overlooked in imbalanced datasets. Specifically, the loss function for TCAD-Net, $\mathcal{L}_{\text{TCAD}}$, is formulated as follows:

$$\mathcal{L}_{\text{TCAD}} = \begin{cases} -\alpha(1-\hat{y})^{\gamma} * ln(\hat{y}) & if \quad y = 1 \\ -(1-\alpha)(\hat{y})^{\gamma} * ln(1-\hat{y}) & if \quad y = 0 \end{cases} \tag{15}$$

where $(1-\hat{y})^{\gamma}$ represents the weight factor, γ is an adjustable multiplier; $\alpha \in [0, 1]$ is a factor used to balance normal samples and anomaly samples. If the sample belongs to a normal sample ($y = 0$), then $\hat{y}$ is close to 0, so $(\hat{y})^{\gamma}$ also close to 0, thereby reducing the influence of easily classified samples on TCAD-Net.

IV. Experimental Validation

This section first introduces the datasets, performance metrics, and parameter settings for TCAD-Net. Subsequently, the proposed TCAD-Net is compared with other anomaly data discovery methods.

A. Datasets

The experiment utilizes eight large-scale benchmark datasets, namely Bank [36], CelebA [37], and Fraud [38] from the consumer domain, Census [39], Shuttle [40], and Backdoor [41] from the electronic information domain, as well as Thyroid [42] and Mammography [40] from the medical domain. Note that anomalous data is defined based on the standards of their respective fields, and the features in each dataset have different dimensions and units. This work follows the standardization methods used in the comparison algorithms DevNet [31] and FEWAE [33] to eliminate the influence of different dimensions on the detection results, remove duplicate data, and handle missing values. The detailed parameter settings of the eight large-scale benchmark datasets are presented in Table III, where "Size" indicates the feature size of the samples in each dataset, "Total" represents the total number of samples, and f_1 represents the percentage of anomalous samples in the training data.

Specifically, the Bank dataset [36] originates from a telephone marketing campaign conducted by a Portuguese banking institution, where anomalies are successful cases in telemarketing (approximately 0.65% of records). The Fraud dataset [38] comes from fraudulent credit card transactions, where the anomalous samples are fraudulent transactions (approximately 6.10% of records). CelebA [37] is sourced from a large-scale image dataset containing over 200K celebrity images, with bald individuals considered as anomalous samples (accounting for approximately 1% of records). Census

TABLE III: Detailed parameter settings of eight large-scale benchmark datasets.

Dataset	Size	Total	f_1	Class	Numbers	Field
Bank	20	41188	0.10%	Normal Anomaly	36548 4640	Consumption
CelebA	39	202599	0.02%	Normal Anomaly	198052 4547	Consumption
Census	500	299285	0.01%	Normal Anomaly	280717 18568	Electronic
Fraud	29	284807	0.01%	Normal Anomaly	284315 492	Consumption
Shuttle	9	49097	0.08%	Normal Anomaly	45586 3511	Electronic
Thyroid	21	7200	0.55%	Normal Anomaly	6666 534	Medical
Backdoor	42	95329	0.04%	Normal Anomaly	93000 2329	Electronic
Mammography	6	11183	0.34%	Normal Anomaly	10923 260	Medical

[39] is derived from the U.S. Census Bureau's database, where individuals with an annual income exceeding $50,000 are defined as anomalous samples (approximately 0.16% of records). The Shuttle dataset [40] is a multi-class classification dataset in the aerospace domain, where the five smallest classes (2, 3, 5, 6, 7) are merged into an anomaly class (approximately 7% of the records). Backdoor [41] is a dataset for intrusion detection in communication networks, where detected backdoor attacks are considered anomalies, and the "normal" class is extracted from the UNSW-NB15 [41]. In the Thyroid [42] dataset, anomalous data refers to patients diagnosed with hypothyroidism (accounting for approximately 5.62% of records). In the Mammography dataset, anomalous data includes a minority diagnosed with calcifications (accounting for about 2.32% of the records).

This work draws on the methods used in other weakly supervised anomaly detection research [33], [34], applying feature encoding and normalization [43], [44] during the data processing phase. This approach helps minimize significant differences in value ranges between different features, reducing their negative impact on the model's accuracy and generalization performance.Before model training, each dataset is divided into training and testing sets, with 80% of the total samples designated for the training set and 20% for the testing set. To simulate the extreme sample imbalance encountered in real-world environments, 30 outlier samples are randomly retained in the training set. Additionally, contaminated outlier samples are added to the normal samples in the training set, maintaining the noise sample level at 2% of the normal samples to simulate the presence of noise in the data.

B. Performance Metrics

PR-AUC (Area under the Precision-Recall Curve) and ROC-AUC (Area under the Receiver Operating Characteristic Curve) are the primary evaluation metrics used to assess the effectiveness of weakly supervised anomaly detection methods [31], [33]. Additionally, this study introduces the F1-Score to further validate the accuracy and reliability of the model. PR-AUC represents the area under the Precision-Recall (PR) curve, where the horizontal axis is Recall and the vertical axis is Precision. The PR curve is particularly useful for evaluating the performance of classification models in situations with imbalanced samples. A larger PR-AUC value indicates better model performance. Similarly, ROC-AUC represents the area under the ROC curve, which depicts the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR). A larger ROC-AUC value indicates that the model achieves higher TPR across varying thresholds of FPR, reflecting superior overall performance. The F1-score, in contrast, provides a comprehensive evaluation by combining precision and recall into a single metric through their harmonic mean.

Note that the value range of PR-AUC and ROC-AUC is between $[0, 1]$. However, unlike ROC-AUC, the PR-AUC of a model that randomly guesses the results of anomaly data discovery will tend towards 0. Therefore, as an evaluation metric, PR-AUC is stricter than ROC-AUC because it focuses more on the model's detection performance for the anomaly class. Consequently, in assessing the effectiveness of weakly supervised anomaly discovery models, PR-AUC is often more important and applicable than ROC-AUC. Moreover, integrating PR-AUC, ROC-AUC, and F1 scores enables the establishment of a multidimensional and in-depth evaluation framework for assessing model effectiveness. This score is used to assess whether the model has achieved a satisfactory balance between precision and recall in detecting anomalous samples.

C. Parameter Settings for TCAD-Net

By default, the TFB in TCAD-Net consists of $M = 8$ identical sub-layers, while the CSB is composed of $N = 7$ sub-layers. Additionally, ablation studies will demonstrate the impact of M and N on the performance of TCAD-Net. The number of heads h in the multi-head attention layer of TFB is set as a factor of the input sample feature dimension to efficiently allocate computational resources and capture information at various levels. The input, output, convolution kernel size, stride, and padding of each sub-layer in CSB are detailed in Table IV, where B denotes the batch size, D represents the dimension of the input features, and S refers to the dimensionality of output features. Note that when average pooling is applied to the input, the stride is set to 2; otherwise, it is set to 1 to maintain stable feature extraction performance. The soft thresholding weight $\Phi = 0.02$ in the CSB. Furthermore, the loss function for TCAD-Net, $\mathcal{L}_{\text{TCAD}}$, is configured with two key hyperparameters: the balance factor $\alpha = 0.85$ and the weighting factor $\gamma = 2$. Each training set undergoes 200 epochs of training with a batch size of 512. Validation results at the end of each training phase are used to assess the model's generalization ability and to help decide whether to retain the trained model.

TCAD-Net is implemented using PyTorch, while the ten competing methods for TCAD-Net are obtained from their respective papers and constructed using either Keras or PyTorch. Experimental comparisons are conducted using the PyTorch 1.7.1 framework on a machine equipped with an NVIDIA GeForce RTX 5000 GPU or an Intel(R) Xeon(R) 5218 CPU, utilizing PyCharm 2022.3.2 (Community Edition).

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TABLE IV: Parameter settings for each sub-layer of CSB.

Layer	Input	Output	Kernel Size	Stride	Padding
1	$B \times 1 \times D$	$B \times 32 \times S$	3	2	1
2	$B \times 32 \times S$	$B \times 64 \times S$	3	1	1
3	$B \times 64 \times S$	$B \times 64 \times S$	3	1	1
4	$B \times 64 \times S$	$B \times 64 \times S$	3	1	1
5	$B \times 64 \times S$	$B \times 64 \times S$	3	1	1
6	$B \times 64 \times S$	$B \times 128 \times S$	3	1	1
7	$B \times 128 \times S$	$B \times 64 \times S$	3	1	1

D. Experimental Results

This subsection compares the proposed TCAD-Net with ten competitive methods on eight datasets, namely iForest [23], Copod [27], Deep-iForest [26], Ecod [28], FSNet$^+$ [29], REPEN$^+$ [30], DevNet [31], Deep-SAD [32], FEWAE [33], and Dual-MGAN [34]. Notably, FSNet$^+$ and REPEN$^+$ are enhanced versions of FSNet and REPEN, respectively, with improvements made to the noise addition methods in Deep-SAD to ensure consistency across each method. As shown in Table V, the last column summarizes the average performance of each method, while the last row displays the results of our proposed TCAD-Net, comparing it with the second best-performing method. At the same time, the F1-Scores for five datasets in the consumer domain including Bank, Fraud, and Backdoor are computed to validate the classification detection performance of TCAD-Net.

Comparing the results in Table V, it is evident that TCAD-Net outperforms the competitive methods across all eight datasets. Specifically, in terms of the average PR-AUC metric, TCAD-Net surpasses Dual-MGAN by 17.96%, FEWAE by 13.81%, Deep-SAD by 11.79%, DevNet by 11.74%, REPEN by 29.07%, FSNet by 36.40%, Ecod by 34.52%, Deep-iForest by 43.71%, Copod by 34.89%, and iForest by 39.74%. Similarly, for the ROC-AUC metric, TCAD-Net's average performance exceeds that of Dual-MGAN by 7.61%, FEWAE by 4.51%, Deep-SAD by 6.18%, DevNet by 4.43%, REPEN by 10.73%, FSNet by 17.38%, Ecod by 11.35%, Deep-iForest by 20.17%, Copod by 13.42%, and iForest by 19.89%. Moreover, on the two largest benchmark datasets, Census and Fraud, TCAD-Net outperforms the second-best method by 2.51% and 3.46%, and 9.07% and 0.41% in PR-AUC and ROC-AUC performance, respectively. Similarly, on the benchmark dataset with relatively fewer samples, Thyroid, TCAD-Net demonstrates strong anomaly detection capabilities, outperforming the second-best method (Deep-SAD) by 17.10% in PR-AUC and 3.50% in ROC-AUC performance. As shown in Table VI, TCAD-Net achieves best F1-Score that surpasses all ten comparative methods across several datasets. On average, TCAD-Net's F1-Score is 1.67% better than that of the second-best method (Deep-SAD). This indicates that TCAD-Net can effectively capture both global and local feature representations from limited data. Despite a 2% contamination level of normal samples, TCAD-Net still achieves accurate anomaly detection, highlighting its reliability and stability.

Overall, TCAD-Net achieves superior anomaly detection results compared to ten competing methods, even with a small number of labeled anomalous samples and the presence of noise. This is attributed to the dual-branch FEM design of TCAD-Net, which effectively enhances feature extraction by efficiently capturing both long-range dependencies and local features. This capability makes TCAD-Net a robust tool for a wide range of anomaly detection applications, offering versatility and high precision.

V. Discussions

Subsection V-A examines the sample efficiency of TCAD-Net. Subsection V-B validates the validity of TCAD-Net. Subsection V-C analyzes the complexity of TCAD-Net across eight large-scale benchmark datasets.

A. Sample Efficiency

Sample efficiency refers to a critical metric that measures how effectively a model utilizes training samples by varying the number of labeled anomaly samples available. This experiment evaluates the performance of TCAD-Net and ten competing methods on eight large-scale benchmark datasets by altering the number of labeled anomaly samples. Specifically, the number of available anomaly samples is varied to 5, 15, 30, 60, and 120, while maintaining a 2% contamination level of normal samples, to assess the sample efficiency of each method.

As shown in Fig. 3, the sample efficiency of these anomaly discovery methods generally increases with the number of labeled anomalous samples across the eight public datasets. TCAD-Net consistently exhibits the highest sample efficiency among all anomaly data discovery methods. In other words, TCAD-Net achieves the best anomaly data discovery performance under different quantities of labeled anomalous samples. It is evident that the PR-AUC performance of TCAD-Net improves with the increase in labeled anomalous samples, particularly in the Bank, CelebA, Thyroid, and Census datasets. When the number of labeled anomalous samples increased from 5 to 120, TCAD-Net's performance improved by 25.27%, 25.14%, 69.2%, and 20.46% on these four datasets, respectively. However, in some cases, increasing the number of labeled anomalies does not enhance model performance. For example, the competing methods Dual-MGAN and DevNet show a decreasing trend within a small range on the Census and Mammography datasets as the number of anomalous samples changes. This may be due to significant differences among the added labeled anomalous samples, causing these models to optimize for the characteristics of certain labeled anomalous samples while neglecting others, leading to poor performance when encountering diverse or infrequent anomalies. For the Fraud and Backdoor datasets, the increase in anomalous samples has a minimal impact on the PR-AUC performance of TCAD-Net, with improvements of only 8.2% and 7.07%, respectively. Similar phenomena can be observed in the ROC-AUC performance.

Overall, as the number of anomalous samples increases, TCAD-Net's performance improves, demonstrating efficient sample utilization across different stages. This indicates that TCAD-Net can effectively leverage limited labeled anomalous samples and possesses strong generalization capabilities.

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TABLE V: PR-AUC and ROC-AUC performance comparisons for TCAD-Net and ten competing methods.

Algorithm	Performance of PR-AUC								
	Bank	CelebA	Census	Fraud	Shuttle	Thyroid	Backdoor	Mammography	Avg.
iForest [23]	0.3280	0.0650	0.0760	0.2540	0.9044	0.1660	0.0660	0.2862	0.2682
Copod [27]	0.4170	0.1565	0.0732	0.3542	0.9511	0.1149	0.0056	0.4595	0.3165
Deep-iForest [26]	0.3560	0.0695	0.0501	0.4091	0.5938	0.0286	0.1068	0.2128	0.2283
Ecod [28]	0.4017	0.1511	0.0705	0.3264	0.8770	0.2297	0.0321	0.4730	0.3202
FSNet$^+$ [29]	0.1930	0.0850	0.1930	0.0430	0.9666	0.1160	0.5730	0.2428	0.3016
REPEN$^+$ [30]	0.3300	0.1610	0.1640	0.6740	0.9586	0.0930	0.1160	0.5021	0.3748
DevNet [31]	0.3810	0.2790	0.3210	0.6900	0.9591	0.2740	0.8830	0.5981	0.5482
Deep-SAD [32]	0.4563	0.1537	0.3123	0.5767	0.9520	0.4933	0.8808	0.5560	0.5476
FEWAE [33]	0.4292	0.2283	0.2837	0.6920	0.9750	0.3145	0.7018	0.5953	0.5275
Dual-MGAN [34]	0.3938	0.2140	0.2392	0.7227	0.9725	0.1650	0.6370	0.5438	0.4860
TCAD-Net(Ours)	**0.6192** (16.29%↑)	**0.3149** (3.59%↑)	**0.3461** (2.51%↑)	**0.8134** (9.07%↑)	**0.9824** (0.74%↑)	**0.6643** (17.10%↑)	**0.9043** (2.13%↑)	**0.6788** (8.07%↑)	**0.6656** (11.74%↑)

Algorithm	Performance of ROC-AUC								
	Bank	CelebA	Census	Fraud	Shuttle	Thyroid	Backdoor	Mammography	Avg.
iForest [23]	0.7310	0.6980	0.6240	0.9530	0.9401	0.6880	0.7520	0.6346	0.7526
Copod [27]	0.7971	0.7505	0.6720	0.9448	0.9762	0.6929	0.7134	0.9272	0.8093
Deep-iForest [26]	0.7292	0.6572	0.5870	0.9627	0.9728	0.4910	0.7846	0.7496	0.7418
Ecod [28]	0.7952	0.7572	0.6575	0.9456	0.9813	0.7485	0.8246	0.9302	0.8300
FSNet$^+$ [29]	0.6230	0.8080	0.7320	0.7340	0.9816	0.5640	0.9280	0.8510	0.7777
REPEN$^+$ [30]	0.7230	0.8940	0.7940	0.9720	0.9657	0.5800	0.8780	0.9473	0.8443
DevNet [31]	0.8070	0.9510	0.8280	0.9800	0.9762	0.7830	0.9690	0.9632	0.9072
Deep-SAD [32]	0.8337	0.7962	0.7993	0.9464	0.9834	0.8525	0.9695	0.9370	0.8898
FEWAE [33]	0.8572	0.9364	0.8274	0.9820	0.9830	0.7445	0.9603	0.9608	0.9065
Dual-MGAN [34]	0.7832	0.8733	0.8361	0.9667	0.9753	0.6733	0.9492	0.9459	0.8754
TCAD-Net(Ours)	**0.9074** (5.02%↑)	**0.9549** (0.39%↑)	**0.8707** (3.46%↑)	**0.9861** (0.41%↑)	**0.9890** (0.56%↑)	**0.8875** (3.50%↑)	**0.9884** (1.89%↑)	**0.9640** (0.08%↑)	**0.9515** (4.43%↑)

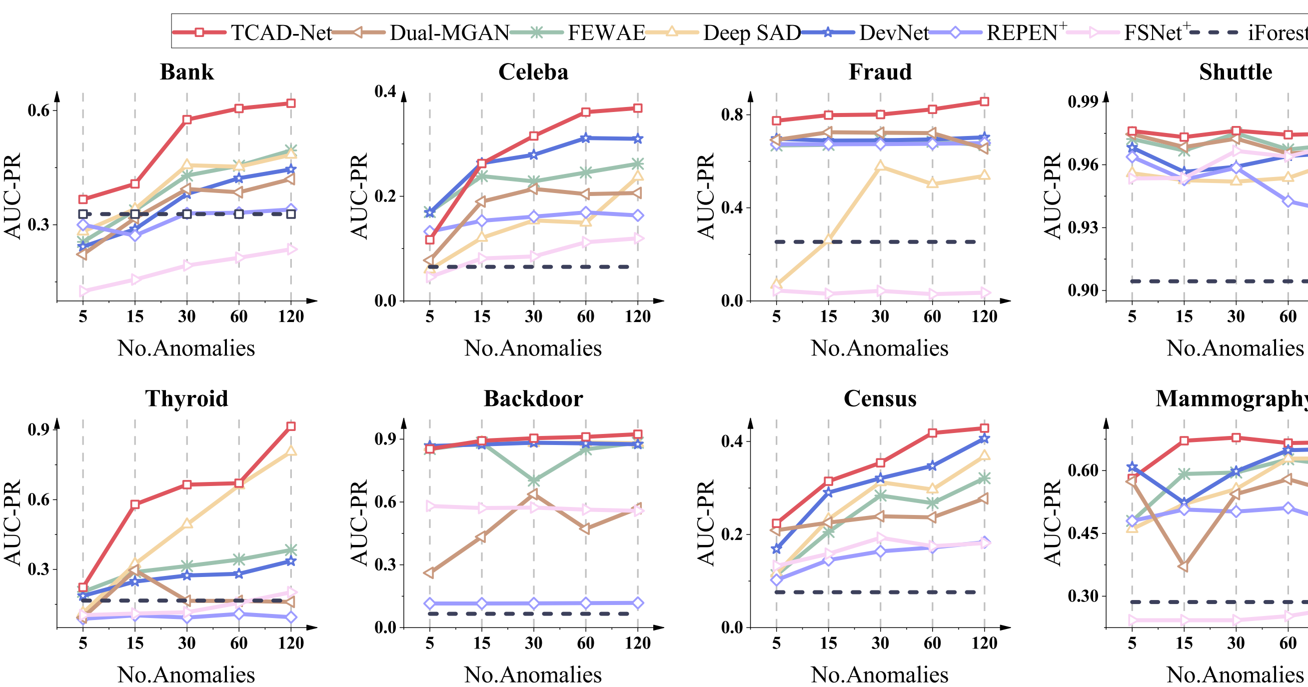

Fig. 3: PR-AUC performance comparisons for eight methods under different numbers of labeled abnormal samples (No. Anomalies).

B. Effectiveness of Dual-branch Design in FEM for TCAD-Net

This subsection demonstrates the effectiveness and validity of the dual-branch FEM in TCAD-Net for integrating long-range dependencies and local features by constructing variants that contain only TFB or CSB. These variants are referred to as TFB-Net and CSB-Net, respectively. In this experiment, the number of anomaly samples is set to 30 and the contamination

TABLE VI: F1-Score performance comparisons for TCAD-Net and ten competing methods.

Algorithm	Performance of F1-Score					
	Bank	CelebA	Fraud	Backdoor	Mammography	Avg.
iForest	0.3573	0.2128	0.2321	0.1328	0.1319	0.2262
Copod	0.4029	0.1512	0.0285	0.0947	0.2850	0.1925
Deep-iForest	0.4707	0.1516	0.5059	0.5427	0.5695	0.5201
Ecod	0.4085	0.1464	0.0925	0.0925	0.2864	0.1925
FSNet$^+$	0.3812	0.2195	0.2101	0.2199	0.1012	0.2723
REPEN$^+$	0.3661	0.2251	0.2123	0.2310	0.1011	0.2778
DevNet	0.3774	0.2178	0.2076	0.2271	0.0917	0.2751
Deep-SAD	0.5277	0.2993	0.7104	<u>0.8881</u>	<u>0.6735</u>	<u>0.6523</u>
FEWAE	<u>0.5504</u>	0.2860	0.6585	0.8488	0.4571	0.5535
Dual-MGAN	0.4720	<u>0.3152</u>	<u>0.7143</u>	0.4652	0.2692	0.4472
TCAD-Net	**0.5584**	**0.3712**	**0.8216**	**0.9055**	**0.6881**	**0.6689**

level of normal samples remains at 2%. Simultaneously, the iteration depths of $M = 8$ for TFB-Net and $N = 7$ for CSB-Net are fixed to maintain consistency in the model architecture. By comparing these standalone detection networks on eight large-scale benchmark datasets, the effectiveness of dual-branch design in FEM for TCAD-Net are evaluated.

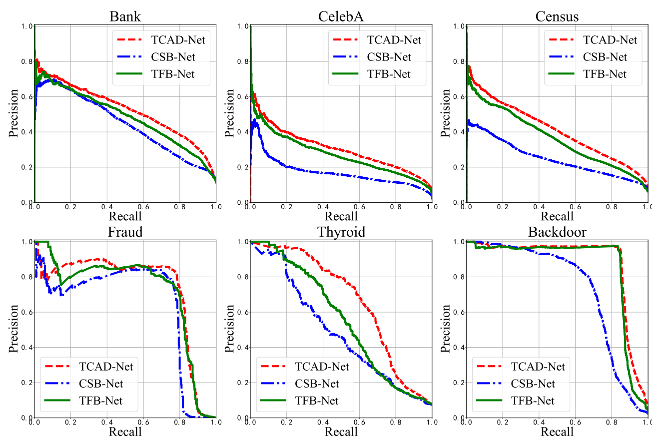

Fig. 4: PR-AUC performance comparisons for TCAD-Net, TFB-Net, CSB-Net arcoss six large-scale benchmark datasets.

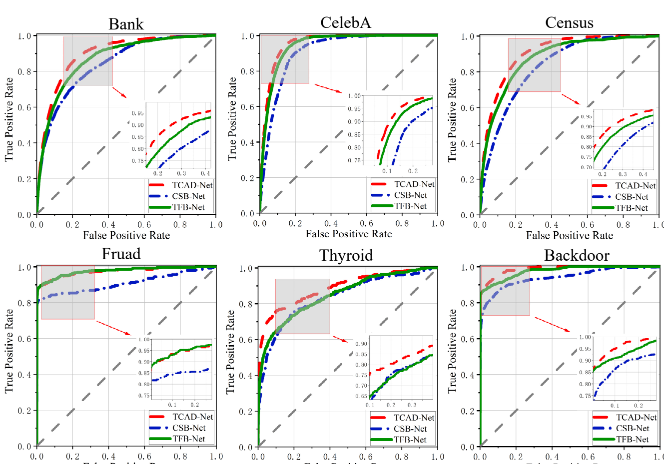

Fig. 5: ROC-AUC performance comparisons for TCAD-Net, TFB-Net, CSB-Net arcoss six large-scale benchmark datasets.

Fig. 4 and Fig. 5 respectively show the PR-AUC curves and ROC-AUC curves of three methods on the Bank, CelebA, Census, Fraud, Thyroid, and Backdoor datasets. The larger the area under the PR-AUC curve and ROC-AUC curve, the better the detection performance. According to these figures, it can be seen that TCAD-Net outperforms TFB-Net and CSB-Net on these datasets. Specifically, TCAD-Net improves the average PR-AUC and ROC-AUC on six datasets by 1.22% and 2.22%, respectively, compared to TFB-Net. Compared to CSB-Net, TCAD-Net improves the average PR-AUC and ROC-AUC by 13.00% and 6.85%, respectively, across the six datasets. Moreover, TFB-Net's PR-AUC and ROC-AUC are also better than those of CSB-Net, exceeding them by 11.79% and 4.46%, respectively. This advantage of TFB-Net may be attributed to its multi-head attention layers and LConvs, which enhance its feature capture capability compared to CSB-Net.

Overall, TCAD-Net demonstrates superior performance on PR-AUC and ROC-AUC curves compared to its variants, i.e., TFB-Net and CSB-Net. This demonstrates that TCAD-Net's FEM effectively integrates the strengths of both TFB-Net and CSB-Net at the feature level, combining local feature information with long-range dependencies. As a result, TCAD-Net is better able to capture subtle variations and overarching patterns in the data, enhancing its ability to handle data diversity and complexity. This leads to more accurate and reliable weakly-supervised anomaly detection.

C. TCAD-Net with Different Branch Depths

This subsection first constructs eight variants of the TCAD-Net by varying the depth of the TFB, $M \in [4, 6, 8, 10]$, and the depth of the CSB, $N \in [1, 3, 5, 7]$, to assess the impact of different branch depths on the performance of TCAD-Net. Note that $M = 8, N = 7$ represents the default configuration. All experimental settings remain consistent, with parameter training rules consistently standardized. During the experiments, each dataset contains 30 labeled anomaly samples while maintaining a fixed contamination level of 2% for the normal samples.

TABLE VII: PR-AUC and ROC-AUC performance comparisons for TCAD-Net with different depths of TFB.

Dataset	PR-AUC				ROC-AUC			
	M=4,N=1	M=6,N=1	M=8,N=1	M=10,N=1	M=4,N=1	M=6,N=1	M=8,N=1	M=10,N=1
Bank	0.5774	0.5794	**0.5836**	0.5827	**0.9021**	0.9011	0.8928	0.8987
Celeba	0.3117	**0.3143**	0.3125	0.3137	0.9539	0.9531	**0.9542**	0.9529
Census	0.3314	0.3419	0.3420	**0.3450**	0.8618	0.8633	0.8641	**0.8642**
Fraud	0.7841	0.8088	0.8067	**0.8114**	0.9852	0.9857	0.9850	**0.9863**
Shuttle	0.9711	0.9711	0.9699	**0.9737**	0.9797	0.9800	**0.9800**	0.9798
Thyroid	0.6474	0.5692	**0.6629**	0.6358	0.8852	0.8583	**0.8865**	0.8587
Backdoor	0.9006	**0.9037**	0.8944	0.8989	0.9859	**0.9872**	0.9859	0.9870
Mammography	0.6749	0.6434	**0.6782**	0.6426	**0.9585**	0.9562	0.9576	0.9559
Avg.	0.6498	0.6415	**0.6563**	0.6505	0.9390	0.9356	**0.9383**	0.9354

TABLE VIII: PR-AUC and ROC-AUC performance comparisons for TCAD-Net with different depths of CSB.

Dataset	PR-AUC				ROC-AUC			
	M=8,N=1	M=8,N=3	M=8,N=5	M=8,N=7	M=8,N=1	M=8,N=3	M=8,N=5	M=8,N=7
Bank	0.5836	0.5782	0.5768	**0.6192**	0.8928	0.8954	0.9012	**0.9074**
Celeba	0.3125	0.3099	0.3140	**0.3149**	0.9542	0.9527	0.9534	**0.9549**
census	0.3420	0.3153	0.3363	**0.3461**	0.8641	0.8617	0.8604	**0.8707**
Fraud	0.8067	0.8116	0.8092	**0.8134**	0.9850	0.9849	0.9850	**0.9861**
Shuttle	0.9699	0.9725	0.9748	**0.9824**	0.9800	0.9786	0.9798	**0.9890**
Thyroid	0.6629	0.6627	0.6614	**0.6643**	0.8865	0.8798	0.8771	**0.8875**
Backdoor	0.8944	0.8980	0.9034	**0.9043**	0.9859	0.9864	0.9874	**0.9884**
Mammography	0.6782	0.6728	0.6724	**0.6788**	0.9576	0.9476	0.9616	**0.9640**
Avg.	0.6563	0.6526	0.6560	**0.6654**	0.9383	0.9359	0.9382	**0.9435**

Table. VII illustrates the performance variations of TCAD-Net in terms of PR-AUC and ROC-AUC across eight large-scale benchmark datasets when $N = 1$ and $M \in [4, 6, 8, 10]$. It is evident that TCAD-Net achieves optimal performance, with average PR-AUC and ROC-AUC values of 0.6563 and 0.9383, respectively, when $M = 8$ and $N = 1$. Consequently, Table. VIII fixes the depth of the TFB at $M = 8$ and varies the depth of the CSB with $N \in [1, 3, 5, 7]$. The results indicate that when the number of layers in the TFB and CSB is set to $M = 8$ and $N = 7$, respectively, TCAD-Net achieves optimal PR-AUC and ROC-AUC values across eight datasets. This provides a strong basis and valuable reference for parameter configuration of the TCAD-Net model in application scenarios within the consumer domain.

D. Robustness Analysis of TCAD-Net

To investigate the performance of TCAD-Net in complex interference environments, this experiment focuses on evaluating its robustness under noise influence. By introducing uniform noise and Gaussian noise, we simulate various interference scenarios that may occur in real-world applications. This approach enables a precise assessment of TCAD-Net's stability and reliability across different noise intensities and distribution characteristics. Specifically, uniform noise refers to random noise uniformly distributed within a given range, while Gaussian noise follows a normal distribution. Additionally, the proportion of anomaly samples in the eight large-scale datasets remains constant, with the level of contamination (noise) controlled to 2% of the normal samples.

TABLE IX: Robustness analysis of TCAD-Net on different datasets.

Dataset	Gaussian Noise			Uniform Noise		
	PR-AUC	ROC-AUC	F1-Score	PR-AUC	ROC-AUC	F1-Score
Bank	0.5867	0.9001	0.5574	0.5830	0.8908	0.5434
CelebA	0.3102	0.9545	0.3644	0.2549	0.9353	0.3123
Census	0.3285	0.8666	0.3659	0.3444	0.8466	0.3715
Fraud	0.7990	0.9852	0.7708	0.7847	0.9836	0.8095
Shuttle	0.9757	0.9790	0.9704	0.9782	0.9870	0.9740
Thyroid	0.6029	0.8668	0.6262	0.6579	0.8853	0.6373
Backdoor	0.8864	0.9820	0.9089	0.9029	0.9857	0.9089
Mammography	0.6637	0.9599	0.6154	0.7119	0.9596	0.6486
Avg.	0.6441	0.9368	0.6474	0.6522	0.9342	0.6507

Table IX presents the robustness evaluation results of TCAD-Net. In this experiment, TCAD-Net demonstrates accurate anomaly detection performance on eight datasets under two different noise environments: Gaussian distribution noise and uniform distribution noise. Notably, under uniform noise, the PR-AUC for the Mammography dataset improved by 3.31% compared to the results in Table V. These findings further demonstrate the robustness and generalization capability of TCAD-Net in the presence of different noise interferences.

E. Complexity of TCAD-Net

Due to variations in feature dimensions among samples, they can affect the complexity of TCAD-Net. Therefore, the Floating Point Operations (FLOPs), Parameters (Params), and size of TCAD-Net across eight large-scale datasets are calculated. TCAD-Net consists of two main modules, FEM and ASGM, with a spatial complexity of $\boldsymbol{O}(n^2)$, where n is the input sample feature dimension. Table X shows the analysis results for TCAD-Net arcoss the eight large-scale datasets.

TABLE X: Complexity comparison of TCAD-Net on different datasets.

Dataset	FLOPs (M)	Params (M)	Size (KB)
Bank	2.5626	0.2576	1585
Celeba	5.8584	0.4192	2221
Census	16.7552	0.9517	4318
Fraud	4.9212	0.3527	1957
Shuttle	1.1605	0.1740	1256
Thyroid	2.7687	0.2687	1629
Backdoor	5.9835	0.4389	2299
Mammography	0.7351	0.1468	1149

TABLE XI: Complexity comparison of TCAD-Net and comparison algorithms on different datasets.

Algorithm	Bank		Fraud		Mammography	
	FLOPs(M)	Params(M)	FLOPs(M)	Params(M)	FLOPs(M)	Params(M)
FSNet	0.0025	0.01	0.0027	0.02	0.0023	0.01
REPEN	0.0184	0.07	0.0196	0.07	0.0166	0.06
Devnet	0.8272	0.1256	0.6556	0.1124	0.5360	0.1032
Deep-SAD	0.3146	0.0847	0.2849	0.0856	0.3316	0.0832
FEWAE	1.8218	1.8399	0.7069	0.2733	0.7385	0.1765
Dual-MGAN	0.1450	0.0730	0.0790	0.0400	0.0330	0.0170
TCAD-Net	2.5626	0.2576	4.9212	0.3527	0.7351	0.1468

Combining Table V and Table X, it can be observed that as the number of sample features increases, TCAD-Net's complexity rises from 0.7351M FLOPs to 16.7552M FLOPs; the model's learnable parameters increase from 0.1740M to 0.9517M; and its size expands from 1149KB to 4318KB. This indicates a positive correlation between model complexity and sample features. Table XI presents a comparison of the FLOPs and Params between TCAD-Net and other baseline algorithms on the Bank, Fraud, and Mammography datasets. As shown in the table, despite the dual-branch architecture employed in TCAD-Net's design, its FLOPs and Params remain under 20M and 1M, respectively. This indicates that even when handling larger-scale data and more complex features, TCAD-Net can efficiently run on resource-constrained edge devices [45], achieving precise weakly-supervised anomaly detection while maintaining low memory usage and computational demands.

VI. Conclusions

This paper proposes TCAD-Net, a weakly-supervised anomaly discovery method that combines Transformer and convolution to effectively model local features and long-range dependencies in large-scale data samples. Specifically, TCAD-Net consists of two main components: FEM and ASGM. The FEM includes two crucial feature extraction branches: TFB and CSB. TFB focuses on capturing long-range dependencies of samples, while CSB is designed to extract local features. Therefore, TCAD-Net comprehensively captures both local and global information of samples, thereby enhancing the accuracy and robustness of anomaly discovery. Furthermore, through comparative experiments on large-scale datasets from eight different domains, TCAD-Net demonstrates its effectiveness in training with limited labeled samples, reducing the dependency on labeled data and outperforming other state-of-the-art weakly supervised anomaly discovery methods. In particular, testing on datasets from the consumer electronics domain demonstrates that TCAD-Net can effectively identify anomalies.

Despite this, due to the black-box nature of DL-based networks, further exploration of the interpretability of TCAD-Net remains necessary in future work. In terms of scalability, when applied to large-scale, complex, and dynamic real-world scenarios such as massive data processing in industrial production, the demand for computational resources will continuously increase, and processing speed is likely to decline. Regarding performance stability, in practical scenarios involving significant noise interference, data loss, or extreme operating conditions—such as high-noise data collected from outdated equipment—the feature extraction and detection capabilities may fluctuate, thereby impacting the performance of weakly supervised anomaly detection. Furthermore, there is potential for lightweight optimization of TCAD-Net's model structure to enhance its adaptability and execution efficiency across various scenarios.

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